



demands larger, more complex, and more precise mirrors—further compounding cost and engineering difficulty. It is similar to trying to read smaller and smaller text using the same pair of glasses. If the lens power cannot be changed, the only option is to build a larger, more precise lens system.

Taken together, these challenges suggest that conventional EUV lithography is approaching practical limits in efficiency, scalability and cost. This raises a bigger question for the industry: Is there a way around these limits, or is chip manufacturing approaching a fundamental ceiling?

Breaking down the new approach

One emerging approach changes how the lithography light itself is produced. Instead of relying on plasma-based EUV sources, it uses a laser-plasma accelerator—a compact particle accelerator driven by ultra-short, high-intensity laser pulses. These pulses generate plasma waves that trap electrons and accelerate them to very high energies over just a few millimetres, rather than the much larger footprints associated with conventional accelerator systems. Imagine giving a sprinter such a powerful push at the starting block that they reach full speed in a few steps instead of needing an entire track.

The accelerated electrons then pass through an undulator, a carefully engineered sequence of alternating magnets. As the electrons oscillate through this magnetic structure, they emit highly coherent EUV or soft X-ray light in a free-electron-laser-style architecture. Unlike conventional EUV sources, this light is tunable, allowing the wavelength to be better aligned with mirror designs and photoresist response windows used in photolithography.

Unlike EUV systems fixed at 13.5nm, this architecture is not fundamentally wavelength-capped. By

adjusting electron energy and undulator design, shorter wavelengths can be generated, positioning the platform not just for the next node, but potentially for multiple future node generations.

So, basically, instead of being stuck with a single piano key, this system can play across a whole keyboard, choosing the exact note needed for each future generation of chip design.

This coherence is one of the most important differences from current systems. Because the emitted light behaves more like a laser beam than an incoherent plasma source, the optical design can eliminate large light-collection optics and use a more compact mirror pathway. The result is lower delivery loss, improved usable brightness at the wafer plane, and the potential for faster wafer exposure. It is the difference between using a torch that shines in one focused direction versus a lantern that scatters light everywhere.

Although the source and beam-delivery path are new, the approach is designed to preserve as much of the downstream scanner architecture as possible. Most notably, the wafer-scanning and positioning subsystems, the precision mechanisms responsible for overlay accuracy, and pattern placement, can remain largely unchanged. This improves the technology's potential compatibility with existing fab workflows, even though full process integration and requalification would still be substantial industrial tasks.

The shorter usable wavelength range also opens a path to continued scaling beyond current EUV approaches, potentially reducing

dependence on increasingly complex multi-patterning strategies. Importantly, this advantage is not solely about wavelength; it also depends on resist chemistry, defect control, and overlay precision, all of which remain critical to achieving future sub-2nm-class process nodes.

Beyond lithography, the same accelerated electrons can also generate X-rays before entering the undulator. These X-rays could support advanced inspection, metrology, and 3D imaging of multilayer chip structures, adding potential value in semiconductor quality control. It is a bit like getting both the printing tool and the inspection camera from the same machine.

From lab to fab

The accelerator platform is compact enough to fit within a standard shipping container-sized footprint and consists of three primary stages. First, an infrared laser generates ultra-short pulses that create plasma in a gas medium, accelerating electrons. Second, magnetic beamline systems shape and stabilise the electron bunches across space, timing, and energy spread. Finally, the electrons pass through the undulator to generate coherent EUV or X-ray light, which is then directed toward the scanner using high-reflectivity mirrors. Throughout the system, thermal management, vibration isolation, and stabilised mirrors keep the beam steady, ensuring precise chip patterning.

Although the light source is new, the wafer-scanning system can remain largely unchanged. The goal is to integrate the system with existing machines with minimal

Next Technical Milestones

- Ongoing system development with major work still ahead
- Near-term focus: Deploy the system for space-related testing
- Next step: Expand into medical applications and build dedicated systems
- Long-term goal (within five years): Move into semiconductor lithography
- Use validated subsystems that meet strict industry standards
- Ensure high levels of quality, reliability, and efficiency for manufacturing use